

NITDGP/BTECH/Reg/Even/2025-26

NATIONAL INSTITUTE OF TECHNOLOGY DURGAPUR
Even Semester Mid-term Examination, 2025-26

Course Code: CEC404

Course Name: Environmental Engineering

Question Paper No.: NITDGP/CEC404/1

Full Marks: 25

Time: 90 Mins

Date of Exam: 26/02/2026

Instructions: Answer Q.5 and any three of the rest questions.

Question No.	Body of the Question	Marks	Mapped CO										
1	Describe in brief any two methods of sinking tube wells with neat sketches of the setups used in them.	2+2	CO1										
2	What are the features of the tubewell used to tap water simultaneously from multiple aquifers? Discuss the problems faced by the critical component used in that tubewell and their solutions.	2+2	CO1										
3	Show the features of a dry river intake structure with a neat sketch. Compare its suitability with a submerged intake structure.	3+1	CO1										
4	What do you understand by the process of well development? What is its importance?	3+1	CO1										
5	(a) A 40cm. dia. well is operating in an unconfined aquifer having coeff. of permeability = 0.0673 m/min. After 24 hours of pumping the water level in a test well at 100m. is lowered by 0.63m. & in another well 25m. away, the drawdown is 1.31m. The coeff. of transmissibility = 1.6825 m ² /min. (i) What is the discharge at the main well? ✓ (ii) If the operating conditions remain the same, what will be the maximum discharge of the well? ✓.	3+4	CO2										
	(b) A test series of water samples to detect the presence of coliform organisms is as follows: 10 ml. – 5/10; 1 ml. – 6/10; 0.1 ml. – 9/10 negative. Comment on the bacteriological quality of water.	2											
	(c) Population data for a typical town is given below: <table border="1"><tr><td>Year</td><td>1971</td><td>1981</td><td>1991</td><td>2001</td></tr><tr><td>Population</td><td>13000</td><td>16500</td><td>18000</td><td>24500</td></tr></table> Estimate the expected population of the town in the year 2031. Justify the method adopted by you in calculating the same.	Year	1971	1981	1991	2001	Population	13000	16500	18000	24500	3+1	
Year	1971	1981	1991	2001									
Population	13000	16500	18000	24500									

$$\frac{m}{min} \times m^2$$

$$P(1 + \frac{(1.4)}{100})^n$$

NATIONAL INSTITUTE OF TECHNOLOGY DURGAPUR
Even Semester End-term Examination, 2025-26

NITDGP/BTECH/Reg/Even/2025-26

Course Code: CEC404

Course Name: Environmental Engineering

Question Paper No.: NITDGP/CEC404/1

Full Marks: 60

Time: 3 Hours

Date of Exam: 27/04/2026

Instructions: Answer any six questions covering all three course outcomes, CO1, CO2 & CO3.

Question No.	Body of the Question	Marks	Mapped CO																
1	(a) Describe the test for the determination of coagulant dosage. Why is alkalinity important in coagulation? (b) Give flowsheets of water treatment processes to be employed for each of the following types of sources of raw water: (i) hilly stretch of a river, & (ii) the flat stretch approaching the confluence with sea. Give reasons for adopting these arrangements.	5+5	CO1																
2	(a) Derive an expression for the correction factor to be used in estimating the flow of water in a closed pipe network by Hardy – Cross Method. (b) What do you understand by the following processes (use sketches, if necessary): (i) Backwashing of filter; (ii) Break point chlorination.	4+6	CO1																
3	(a) Define BOD and explain the difference between first & second stages of it. (b) Describe the determination of BOD by dilution method.	4+6	CO1																
4	(a) Give the flowsheet of a municipal sewage treatment plant operating with a single stage high rate trickling filter. Where does it differ from a two stage arrangement of the same biological unit? (b) What is the importance of recirculation in a typical biological sewage treatment unit?	6+2+2	CO1																
5	(a) A settling tank is designed to have a surface overflow rate of 32.5 m/d. Determine the overall removal obtained for a water sample with the following particle size distribution. The specific gravity of the particles is 1.2, the dynamic viscosity is 1.027×10^{-2} gm/cm-sec and the density is 0.997 gm/cc. <table border="1"><tr><td>Particle size (mm)</td><td>0.1</td><td>0.08</td><td>0.07</td><td>0.06</td><td>0.04</td><td>0.02</td><td>0.01</td></tr><tr><td>Weight fraction greater than size (%)</td><td>10</td><td>12</td><td>38</td><td>72</td><td>90</td><td>97</td><td>100</td></tr></table> (b) What do you understand by the following: (i) Rising main (ii) Ferrule connection.	Particle size (mm)	0.1	0.08	0.07	0.06	0.04	0.02	0.01	Weight fraction greater than size (%)	10	12	38	72	90	97	100	6+4	CO2
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Weight fraction greater than size (%)	10	12	38	72	90	97	100												
6	(a) Explain the process of self-purification of streams through the development of D.O. sag curve. (b) A stream saturated with D.O. has a flow of $1.3 \text{ m}^3/\text{s}$, BOD_5 of 4 mg/l and rate constant (base 10) of 0.3/day. From a township, it receives an effluent discharge having $\text{BOD}_5 = 25 \text{ mg/l}$, D.O. = 4mg/l and rate constant (base 10) = 0.12/day. The township has a population of 1,20,000 with average water supply @ 135 lpcd. The average velocity of flow of the stream is 0.2m/s. Calculate the D.O. deficit at points 20km and 80km downstream and comment on these values. Assume that the temperature is 20°C throughout with the saturation D.O. = 9.17mg/l.	4+6	CO1 CO2																

Course Outcomes

CO1: Apply knowledge of water supply & wastewater engineering for design solutions.

7	A water treatment plant treats 40×10^6 lpd of water using coagulation – sedimentation tanks. The dosage of filter alum is 20mg/l. If the alkalinity of raw water is equivalent to 4.5mg/l of CaCO_3 , determine the quantity of filter alum and quick lime (containing 80% of CaO) required per month by the plant. Take molecular weights as : $\text{Ca} = 40$; $\text{C} = 12$; $\text{S} = 32$; $\text{O} = 16$; $\text{Al} = 27$ & $\text{H} = 1$.	10	CO2
8	A conventional activated sludge plant treats settled domestic sewage for the following data : (i) Population = 1,20,000 ; (ii) Water supply = 140lpcd ; (iii) Raw sewage $\text{BOD}_5 = 300\text{mg/l}$; (iv) Effluent BOD_5 required = 15mg/l ; (v) BOD_5 removal efficiency of primary settling tank = 30% ; (vi) $F/M = 0.2$; (vii) $\text{MLSS} = 3000\text{mg/l}$ (viii) $\text{SVI} = 100$; (ix) Air requirement = $100\text{m}^3/\text{day/kg}$ BOD_5 removed. Determine the following : (a) Required process efficiency ; (b) Volume of aeration tank ; (c) HRT ; (d) Volumetric loading ; (e) Return sludge ratio ; (f) Air requirement per minute.	$1+2 \times 4 + 1$	CO3
9	Design an oxidation pond system for the following data : (a) Areal BOD_5 loading = 175kg/ha/d ; (b) Elevation = 600m above MSL ; (c) Range of mean monthly temperature = 30°C (max.) – 10°C (min.) ; (d) Population to be served = 10,000 ; (e) Water supply = 120lpcd ; (f) BOD_5 of sewage influent to the pond = 250mg/l ; (g) Desired effluent $\text{BOD}_5 = 30\text{mg/l}$; (h) Pond removal constant at $20^\circ\text{C} = 0.1/\text{day}$; (i) Sky clearance factor = 60% ; (j) For the ponds, $L/B = 3$.	10	CO3
10	A rectangular sedimentation tank is treating 3×10^6 lpd of water. The size of the tank is $18\text{m} \times 5.5\text{m} \times 3.5\text{m}$. If 90mg/l suspended solids are present in water with average specific gravity of 2.0, determine the following: (a) Average velocity of flow through the tank in cm/min.; (b) HRT in hrs.; (c) Deposition of solids in the tank in kg, assuming 75% removal in the tank; (d) Overflow rate in $\text{m}^3/\text{d/m}^2$.	10	CO3
11	(a) Show the stratification in a conventional anaerobic digester. (b) Design a septic tank and the associated soil absorption system for the following data: (i) Number of users = 150; (ii) Peak discharge of 720lpm; (iii) Soil percolation rate = 20min/cm; (iv) Standard space requirements under appropriate heads (to be specified in the design) : $0.92\text{m}^3/10\text{lpm}$; $0.032\text{m}^3/\text{capita}$ and $0.073\text{m}^3/\text{capita}$; Provide the sketch of sectional elevation of both the units.	$3+4+3$	CO2 CO3